Exploration of Basic Calculus and Linear Regression

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Introduction

- My original focus for our linear algebra project was to create and explore simplistic linear regression models.
- Today, I have 3 experiments that are attempted variations on that theme.

Explanation

- Interested in Computer Science
- Interested in Programming
- Interested in Machine Learning

- Wanted to focus on a more educational than practical implementation

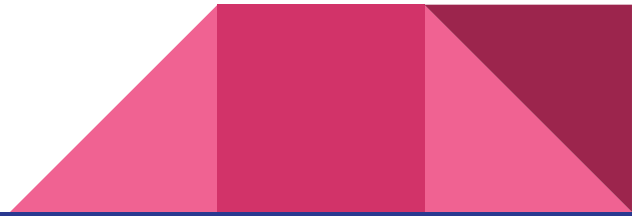


Experiment #1 Intro

- Wanted to make a full fledged linear regression model.
- Haven’t taken multivariate calculus.

General Linear Function

$f(x) = mx + b$



Experiment #1: Approach

Model: f(x)= mx + b

Weights: [m,b]

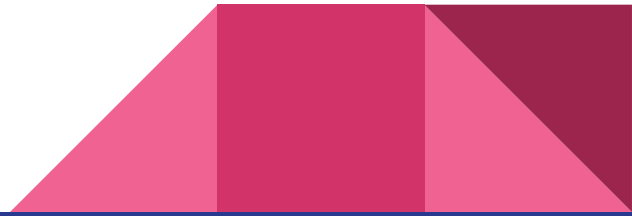
Error Criterion Tested: Sum Squared Error, Mean Squared Error, Root Mean Squared Error, Binary Log Loss

Notes:

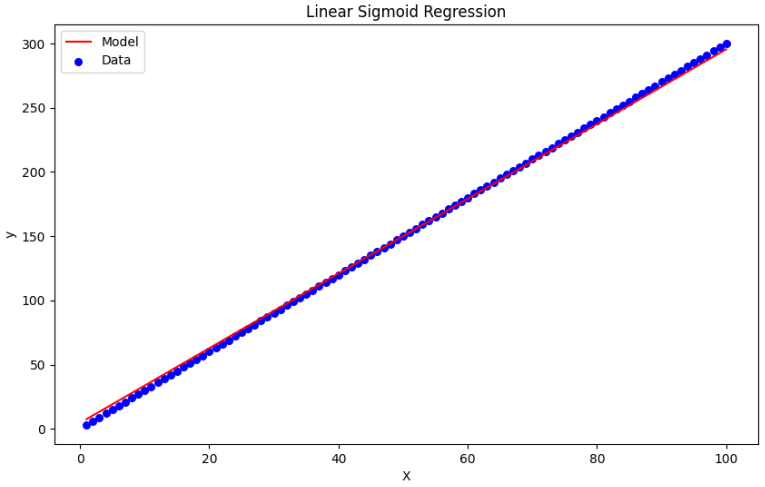
- The data has a single feature, x.
- The X and y data is stored in matrices.
- Tried to reuse the function to create a binary classification model

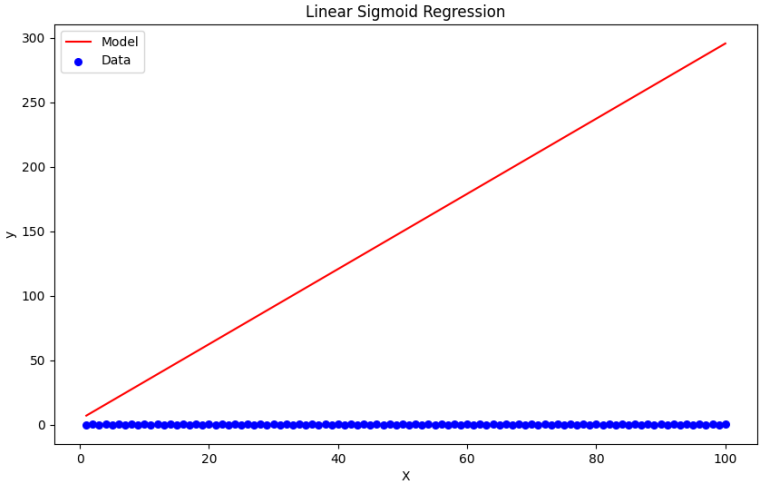
Experiment #1: Challenges

- Storage of weights and data
- Tuning parameters like error threshold and learning rate
- No regularization or feature scaling
- Weight initialization



Experiment #1: Results





Experiment #2: Intro

- Initial model was a failure for many tasks.
- Wanted a non linear model to test those use cases

General Quadratic

$f(x)= A + Bx + Cx^2$

Experiment #2: Approach

Model: $f(x)= A + Bx + Cx^2$

Weights: [A,B,C]

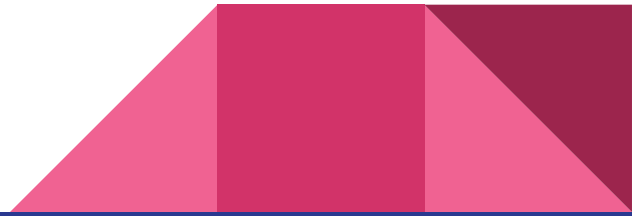
Error Criterion Tested: SSR, MSE, RMSE

Notes:

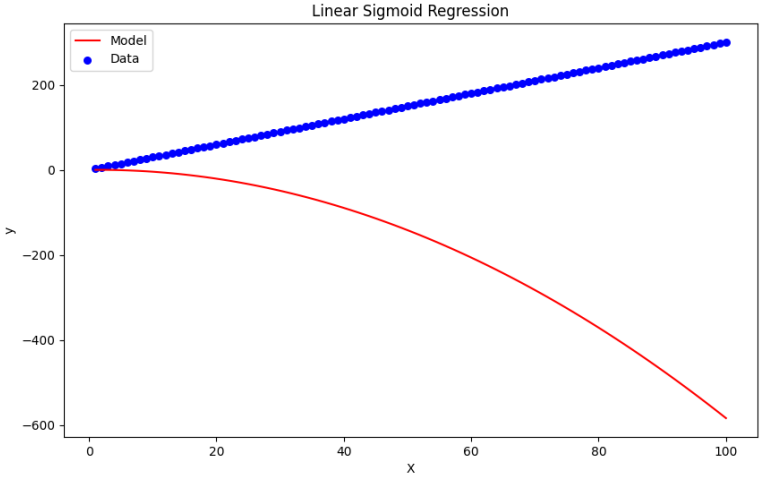
- Used a data standardization technique.

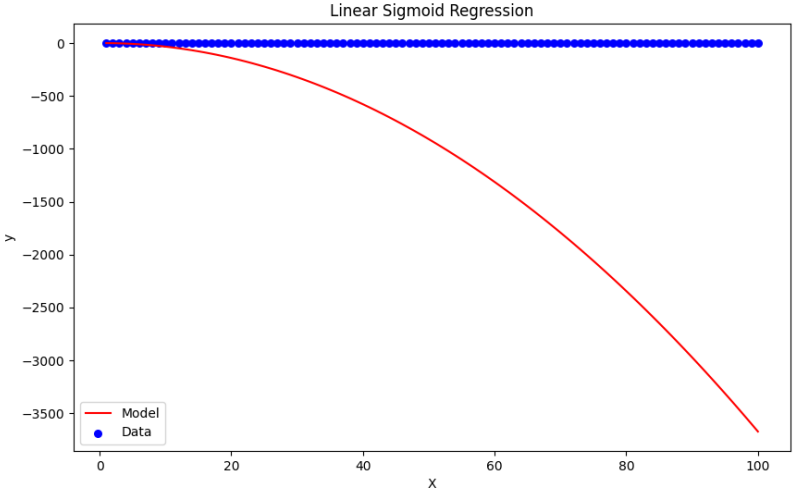
Experiment #2: Challenges

- High growth rate of quadratics makes the training more volatile
- Overfitting
- Poor Extrapolation



Experiment #2 Results





Experiment #3: Intro

- Disappointed in the linear and quadratic model's performance in classification tasks
- Wanted to see if the same issues persisted at high degrees

N-th Degree Polynomial

$f(x)= A + Bx + Cx^2 + Dx^3 + \ldots$

Experiment #3: Approach

Model: $f(x)= A + Bx + Cx^2 + \ldots$

Weights: [A,B,C, …]

Error Criterion Tested: SSR, MSE, RMSE

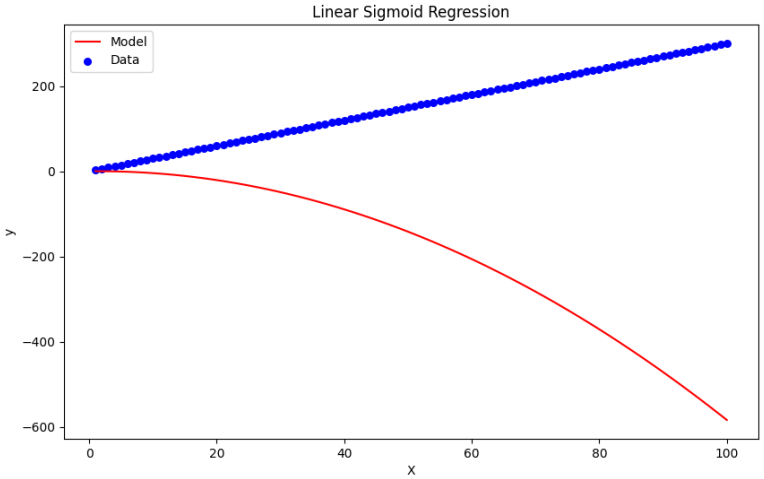
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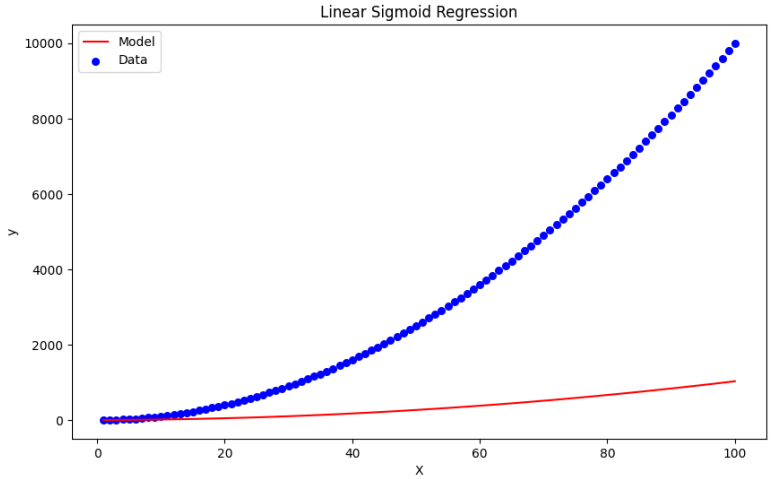
- Used the same data standardization technique.

Experiment #3: Challenges

- High growth rate seen in quadratics only increases
- The starting weights massively impacted accuracy, training time
- Overfitting
- Poor Extrapolation

Experiment #3 Results



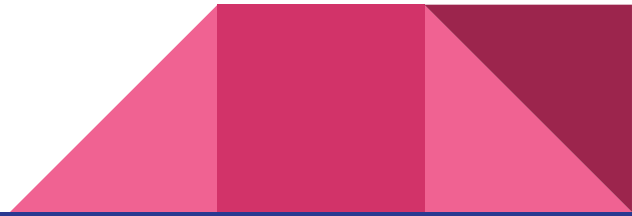


Experiment #4: Intro

- Disappointed in the other models' performance in classification tasks
- Wanted to see if I could create a very simple non-linear model

Connected Linear Functions

Connects f(x) to g(x), h(x) at intersections with sigmoid.

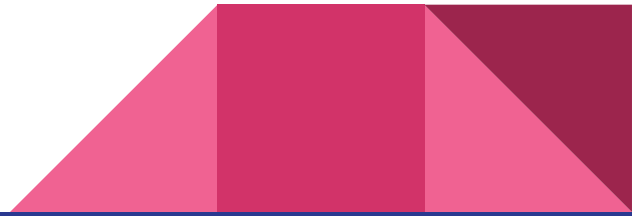


Experiment #4: Approach

Model: f(x)= mx + b

Weights: [m,b] * number of functions

Error Criterion Tested: SSR, MSE, RMSE

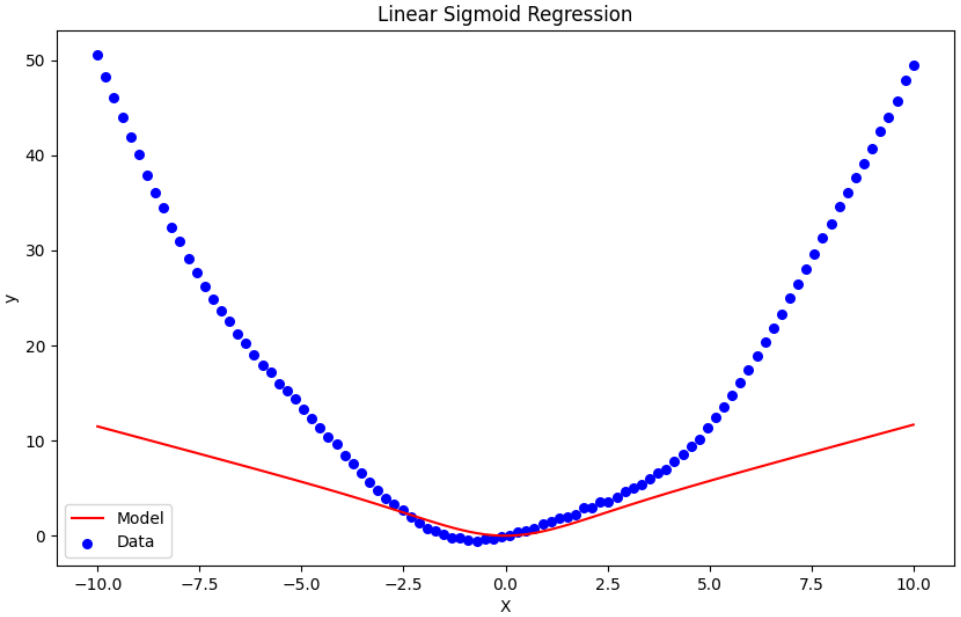


Experiment #4: Challenges

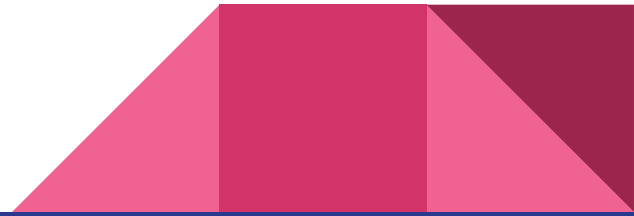
- Learning the sigmoid function and more stable versions
- Figuring out how to distribute the functions over intervals



Experiment #4: Results

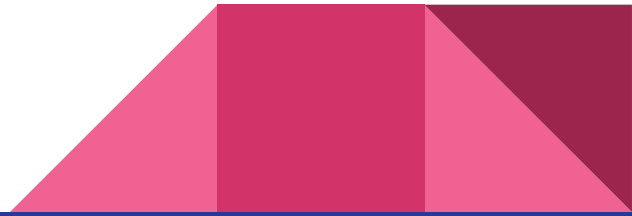


Code Review



Notes from project

- These are not complex models.
- The models are quite dumb.
- Cannot create complicated model with my current skills.
- Calc 3 is definitely a must for future education.
- Implementations are much harder without some of the standard packages.



Next Steps

- Introduce L1,L2 Regularization (Lasso & Ridge) to reduce overfitting.
- Research techniques to stabilize the training of polynomial model.
- Create a regression model based on Taylor Series.
- Review Multivariate Calculus and try to implement a fully fledged multi-feature linear regression model.
- Create simple non-linear NN regression model.

